



MODEL ASSUMPTIONS

3. Efficient local exploration (ELE): For all β if $X' | X \sim K_\beta(X, d_{X'})$ then $V(X') \perp V(X)$

► We will make the following assumptions:

4. **Regularity:** V has third moments with respect to η and π

2. **Stationarity:** Suppose $\mathbf{X}_0 \sim \pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$

- Ensures a proposed swap between components $i, i + \epsilon$ occurs with probability $s_{i,\epsilon}$

$$s_{i,\epsilon} = \mathbb{E}[\alpha_{i,i+\epsilon}(\mathbf{x})], \quad \mathbf{X} \sim \pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$$

► ELLE ensures the energies $V(X_f^n)$ are independent between iterations and chains

▶ Always to study the impact of municipal aid on independent local movements

▶ Note that ELEE gives a model to analyse P_T, but we do assume it holds in practice

► In general P^T is robust to server violations.

Assumptions can be extended to a general path

1. **Non-degeneracy:** Suppose π_β is linear path with energy $V = \log \eta - \log \gamma \neq 0$

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2. **Stationarity:** Suppose $\mathbf{X}_0 \sim \pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$

3. **Efficient local exploration (ELE):** For all β if $X' | X \sim K_\beta(X, dx')$ then $V(X') \perp V(X)$

4. **Regularity:** V has third moments with respect to η and π

- ▶ ELE ensures the energies $V(X_t^n)$ are independent between iterations and chains

- ▶ Ensures a proposed swap between components $i, i + \epsilon$ occurs with probability $s_{i,\epsilon}$

$$s_{i,\epsilon} = \mathbb{E}[\alpha_{i,i+\epsilon}(\mathbf{x})], \quad \mathbf{X} \sim \pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$$

- ▶ Note that ELE gives a model to analyse PT, but we do assume it hold in practice
 - ▶ Allows us to study the impact of communication independent the local move
 - ▶ In general PT is robust to severe violations.
- ▶ Assumptions can be extended to a general path

EMPERICAL JUSTIFICATION OF ELE